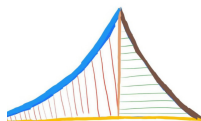


Math for EE

1



G. V. V. Sharma

ABOUT THIS BOOK

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

License: <https://creativecommons.org/licenses/by-sa/3.0/>
and

<https://www.gnu.org/licenses/fdl-1.3.en.html>

First manual appeared in January 2018

First edition published on July 10, 2024

In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

CONTENTS

1	GATE DA 2025	4
1.1	Formulae	4
1.2	Problems	12

1. GATE DA 2025

A. Formulae

1.1.1 Indicator Summation for Thresholding

$$N = \sum_{i=1}^M \sum_{j=1}^N \mathbb{I}(U_{ij}) \quad (1.1.1.1)$$

Solution: The total count of elements in a matrix $\mathbf{U}$ satisfying a threshold c is obtained by summing the indicator function $\mathbb{I}$. The values of $\mathbb{I}$ operate as a step function defined as:

$$\mathbb{I}(U_{ij}) = \begin{cases} 1, & U_{ij} \leq c \\ 0, & U_{ij} > c \end{cases} \quad (1.1.1.2)$$

1.1.2 Geometric Scaling under Dilation

$$A' = (1 + \alpha) A \quad (1.1.2.1)$$

Solution: For a rectangle with length L and width W , Area $A = LW$. If W is scaled by a factor of $(1 + \alpha)$:

$$W' = (1 + \alpha) W \quad (1.1.2.2)$$

Substituting (1.1.2.2) into the area equation:

$$A' = LW' = L[(1 + \alpha) W] = (1 + \alpha) LW = (1 + \alpha) A \quad (1.1.2.3)$$

yielding (1.1.2.1).

1.1.3 Injectivity of Strictly Unimodal Functions

$$x_1 < c < x_2 \quad \text{for} \quad f(x_1) = f(x_2) \quad (1.1.3.1)$$

Solution: If $f(x)$ strictly increases on $[0, c]$ and strictly decreases on $[c, \infty)$, each branch is individually injective (one-to-one). If two distinct points x_1 and x_2 yield the same function value, they cannot lie on the same monotonic branch, meaning the peak c must lie strictly between them.

1.1.4 Exponent Laws

$$\frac{a^u}{a^v} = a^{u-v}, \quad (a^x)^y = a^{xy} \quad (1.1.4.1)$$

1.1.5 Law of Total Expectation (Tower Property) Let X and Y be continuous random variables. The initial integral equations defining the expectations are:

$$g(y) = E[X | Y = y] = \int_{-\infty}^{\infty} x f_{X|Y}(x | y) dx \quad (1.1.5.1)$$

$$E[E[X | Y]] = \int_{-\infty}^{\infty} g(y) f_Y(y) dy \quad (1.1.5.2)$$

$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx \quad (1.1.5.3)$$

1.1.6 Computational Complexity of Back-Substitution For an $n \times n$ upper triangular matrix system $\mathbf{U}\mathbf{x} = \mathbf{b}$, variables are obtained sequentially:

$$x_i = \frac{1}{U_{ii}} \left(b_i - \sum_{j=i+1}^n U_{ij}x_j \right) \quad (1.1.6.1)$$

1.1.7 Eigenvalues of Constant Row-Sum Matrices and System Solvability

$$\mathbf{A}\mathbf{1} = \mathbf{1} \cdot \mathbf{1}, \quad (\mathbf{A} - \mathbf{I})\mathbf{1} = \mathbf{0} \cdot \mathbf{1} \quad (1.1.7.1)$$

Solution: Let $\mathbf{a}_i^\top$ denote the i -th row of $\mathbf{A} \in \mathbb{R}^{n \times n}$. Because the sum of the elements in each row is 1, multiplying the row by the all-ones column vector $\mathbf{1} = (1 \ 1 \ \dots \ 1)^\top$ yields:

$$\mathbf{a}_i^\top \mathbf{1} = 1, \quad \forall i \in \{1, 2, \dots, n\} \quad (1.1.7.2)$$

Grouping these row equations together to reconstruct the full matrix $\mathbf{A}$ gives:

$$\mathbf{A}\mathbf{1} = \begin{pmatrix} \mathbf{a}_1^\top \\ \mathbf{a}_2^\top \\ \vdots \\ \mathbf{a}_n^\top \end{pmatrix} \mathbf{1} = \begin{pmatrix} \mathbf{a}_1^\top \mathbf{1} \\ \mathbf{a}_2^\top \mathbf{1} \\ \vdots \\ \mathbf{a}_n^\top \mathbf{1} \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} = \mathbf{1} = \mathbf{1} \cdot \mathbf{1} \quad (1.1.7.3)$$

This demonstrates that the vector $\mathbf{1}$ is an eigenvector of $\mathbf{A}$ associated with the eigenvalue $\lambda = 1$.

To find the eigenvalue of $\mathbf{A} - \mathbf{I}$ corresponding to the same eigenvector $\mathbf{1}$:

$$(\mathbf{A} - \mathbf{I})\mathbf{1} = \mathbf{A}\mathbf{1} - \mathbf{I}\mathbf{1} = \mathbf{1} - \mathbf{1} = \mathbf{0} = \mathbf{0} \cdot \mathbf{1} \quad (1.1.7.4)$$

Thus, $\mathbf{A} - \mathbf{I}$ possesses an eigenvalue of 0, meaning it is a singular matrix ($|\mathbf{A} - \mathbf{I}| = 0$).

Number of Solutions for Linear Systems:

- For the system $(\mathbf{A} - \mathbf{I})\mathbf{x} = \mathbf{b}$: Because $\mathbf{A} - \mathbf{I}$ is singular ($\text{rank} < n$), the system will have infinitely many solutions if $\mathbf{b}$ is in the column space of $\mathbf{A} - \mathbf{I}$ (i.e., $\text{rank}(\mathbf{A} - \mathbf{I}) = \text{rank}([\mathbf{A} - \mathbf{I} \mid \mathbf{b}])$), or no solution if it is not.
- For the system $\mathbf{A}\mathbf{x} = \mathbf{b}$: The number of solutions depends entirely on the remaining unknown eigenvalues of $\mathbf{A}$. If all other eigenvalues are non-zero ($|\mathbf{A}| \neq 0$), the system has a unique solution. If 0 happens to also be an eigenvalue of $\mathbf{A}$, the matrix is singular, resulting in either no solution or infinitely many solutions

1.1.8 Maclaurin Series of Hyperbolic Sine

$$\sinh(x) = \sum_{k=0}^{\infty} \frac{x^{2k+1}}{(2k+1)!} \quad (1.1.8.1)$$

Solution: Expanding e^x and e^{-x} via Maclaurin series :

$$f(x) = \frac{1}{2} \left(\sum_{n=0}^{\infty} \frac{x^n}{n!} - \sum_{n=0}^{\infty} \frac{(-1)^n x^n}{n!} \right) \quad (1.1.8.2)$$

$$= x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots \quad (1.1.8.3)$$

1.1.9 Relational Algebra Natural Join

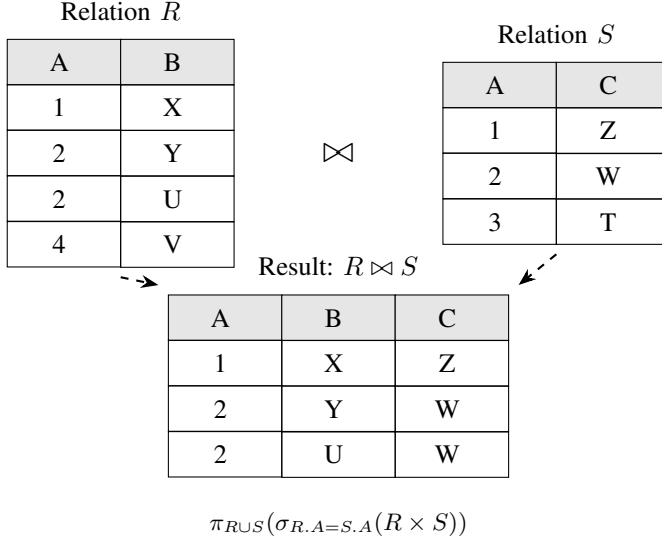


Fig. 1.1.9.1: Pictorial representation of Relational Algebra Natural Join

$$R \bowtie S = \pi_{R \cup S}(\sigma_{R.A=S.A}(R \times S)) \quad (1.1.9.1)$$

Solution: The equation describes the mechanical steps of combining two database tables using relational algebra operators:

- R and S : The two input relations (data tables) being operated on.
- $\bowtie$ (Natural Join): The overarching operator that merges R and S by matching rows with identical values in their common columns, outputting a single combined table with no duplicate columns.
- $R \times S$ (Cartesian Product): The foundational operation that pairs every single row in table R with every single row in table S , generating all possible combinations.
- $\sigma_{R.A=S.A}$ (Selection): The row-filtering step. The σ operator evaluates the Cartesian product and retains only the rows where the value in the shared attribute A matches between both tables.
- $\pi_{R \cup S}$ (Projection): The column-filtering step. The π operator dictates which columns appear in the final result. The subscript $R \cup S$ ensures that all unique

columns from both tables are kept, effectively stripping out the redundant duplicate common column ($S.A$) generated by the Cartesian product.

1.1.10 Linear Probing in Hash Tables

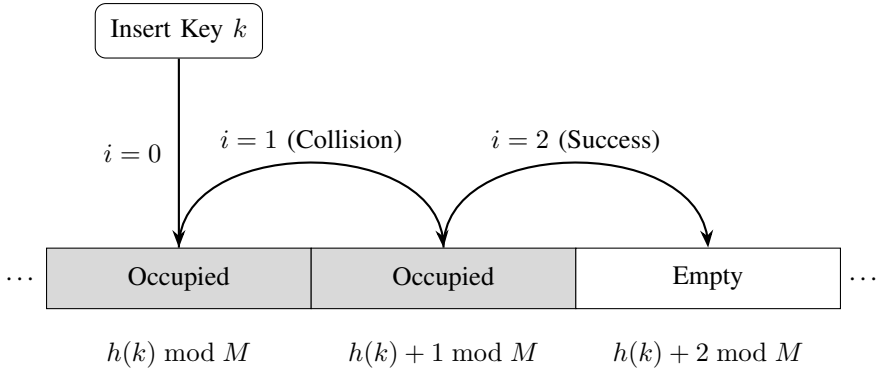


Fig. 1.1.10.1: Pictorial representation of Linear Probing in Hash Tables

$$h(k, i) = (h(k) + i) \pmod{M} \quad (1.1.10.1)$$

The terms in the probing function represent:

- k : The original key or piece of data to be stored.
- $h(k)$: The primary hash function that calculates the initial target index.
- i : The probe attempt number, starting at 0 and incrementing linearly for each collision encountered.
- M : The total allocated size of the hash table.
- $\pmod{M}$: The modulo operation ensuring the probe sequence wraps around to index 0 if the end of the table is reached.
- $h(k, i)$: The final calculated storage index for the i -th probe attempt.

1.1.11 Median of a Continuous Distribution

Let $f_X(x)$ be the probability density function. The median m divides the total probability mass strictly in half:

$$\Pr(X \leq m) = 0.5 \quad (1.1.11.1)$$

$$\int_{-\infty}^m f_X(x) dx = 0.5 \quad (1.1.11.2)$$

Because the integral of $f_X(x)$ from $-\infty$ to m is exactly $F_X(m)$, it follows that:

$$F_X(m) = \int_{-\infty}^m f_X(x) dx = 0.5 \quad (1.1.11.3)$$

1.1.12 Let the random variable Z follow a standard normal distribution.

It follows that

$$E[Z] = 0 \quad (1.1.12.1)$$

$$E[Z^2] = 1 \quad (1.1.12.2)$$

Show that for

$$X = aZ + b \quad (1.1.12.3)$$

It follows that

$$E[X] = aE[Z] + b \quad (1.1.12.4)$$

$$\text{Var}(X) = a^2 \text{Var}(Z) \quad (1.1.12.5)$$

Solution: The step-by-step derivations for the expectations are:

$$E[X] = E[aZ + b] \quad (1.1.12.6)$$

$$= aE[Z] + b \quad (1.1.12.7)$$

$$= a(0) + b \quad (\text{using (1.1.12.1)}) \quad (1.1.12.8)$$

$$= b \quad (1.1.12.9)$$

Then,

$$X - E[X] = (aZ + b) - b \quad (1.1.12.10)$$

$$= aZ \quad (1.1.12.11)$$

Similarly,

$$E[(X - E[X])Z] = E[aZ \cdot Z] \quad (1.1.12.12)$$

$$= aE[Z^2] \quad (1.1.12.13)$$

$$= a(1) \quad (\text{using (1.1.12.2)}) \quad (1.1.12.14)$$

$$= a \quad (1.1.12.15)$$

Yielding,

$$E[(X - E[X])^2] = E[(aZ)^2] \quad (1.1.12.16)$$

$$= a^2 E[Z^2] \quad (1.1.12.17)$$

$$= a^2(1) \quad (\text{using (1.1.12.2)}) \quad (1.1.12.18)$$

$$= a^2 \quad (1.1.12.19)$$

1.1.13 Tail Probability of Exponential Distribution

$$\Pr(X \geq x) = \int_x^\infty \lambda e^{-\lambda t} dt = e^{-\lambda x} \quad (1.1.13.1)$$

1.1.14 Python List and Set Operations

$$A.\text{update}(B) \implies A \leftarrow A \cup B \quad (1.1.14.1)$$

Solution: Let $A = (a_1, a_2, \dots, a_n)$ and $B = (b_1, b_2, \dots, b_m)$ represent sequences, and let $\leftarrow$ denote in-place modification. The functions map to the following mathematical transformations[cite: 41]:

$$A.\text{extend}(B) \implies A \leftarrow (a_1, \dots, a_n, b_1, \dots, b_m) \quad (1.1.14.2)$$

$$A.\text{append}(B) \implies A \leftarrow (a_1, \dots, a_n, B) \quad (1.1.14.3)$$

$$A.\text{update}(B) \implies A \leftarrow A \cup B \quad (\text{where } A \text{ and } B \text{ are sets}) \quad (1.1.14.4)$$

$$A.\text{insert}(i, B) \implies A \leftarrow (a_1, \dots, a_i, B, a_{i+1}, \dots, a_n) \quad (1.1.14.5)$$

1.1.15 Differentiability Rules

$$(fg)'(c) = f'(c)g(c) + f(c)g'(c) \quad (1.1.15.1)$$

Solution: If f and g are differentiable at c , their sum, difference, and product are strictly differentiable at c . The quotient is differentiable if $g(c) \neq 0$. Composition $f \circ g$ requires f to be differentiable at $g(c)$, which is not universally guaranteed.

1.1.16 Properties of Orthonormal Sets

$$\mathbf{u}_i^\top \mathbf{u}_j = \delta_{ij} \quad (1.1.16.1)$$

Solution: An orthonormal set of vectors in $\mathbb{R}^n$ is unconditionally linearly independent. Orthonormal bases in $\mathbb{R}^n$ are not unique because applying any rotation matrix generates a new valid orthonormal basis.

1.1.17 Binary Search Complexity

$$T(n) = T(n/2) + O(1) \implies T(n) = O(\log n) \quad (1.1.17.1)$$

Solution: Binary search requires $O(1)$ random access to indices, restricting its $O(\log n)$ performance bounds strictly to contiguous arrays, not linked lists.

1.1.18 Matrix Determinant Lemma

$$|\mathbf{I} + \mathbf{u}\mathbf{v}^\top| = 1 + \mathbf{v}^\top \mathbf{u} \quad (1.1.18.1)$$

Solution: For $\mathbf{A} = \mathbf{I}_n + \mathbf{x}\mathbf{x}^\top$ with $\mathbf{x}^\top \mathbf{x} = 1$:

$$|\mathbf{A}| = 1 + \mathbf{x}^\top \mathbf{x} = 2 \neq 0 \quad (1.1.18.2)$$

Because $|\mathbf{A}| \neq 0$, $\mathbf{A}$ is invertible and has full rank. Its eigenvalues are 2 and 1, neither being zero or negative.

1.1.19 Insertion Sort Inversions

$$\text{Swaps} = \text{Number of Inversions} \quad (1.1.19.1)$$

Solution: Insertion sort executes one swap per inverted pair. For the array to be sorted in exactly two swaps, the inserted element x must be strictly smaller than exactly two preceding elements.

1.1.20 Bayes' Theorem for Partitioned Spaces

$$\Pr(B_j | W) = \frac{\Pr(W | B_j) \Pr(B_j)}{\sum_i \Pr(W | B_i) \Pr(B_i)} \quad (1.1.20.1)$$

1.1.21 Asymptotic Radical Limits

$$\lim_{t \rightarrow \infty} \left(\sqrt{t^2 + at} - t \right) = \frac{a}{2} \quad (1.1.21.1)$$

Solution: Multiplying by the conjugate:

$$\lim_{t \rightarrow +\infty} \frac{(t^2 + t) - t^2}{\sqrt{t^2 + t} + t} = \lim_{t \rightarrow +\infty} \frac{1}{\sqrt{1 + \frac{1}{t}} + 1} = 0.5 \quad (1.1.21.2)$$

1.1.22 Least Squares Regression Through Origin

$$w^* = \frac{\sum x_i y_i}{\sum x_i^2} \quad (1.1.22.1)$$

Solution: Minimizing $S(w) = \sum (y_i - wx_i)^2$:

$$\frac{dS}{dw} = -2 \sum x_i (y_i - wx_i) = 0 \implies w^* = \frac{24}{14} \approx 1.714 \quad (1.1.22.2)$$

1.1.23 Bayes Optimal Error Rate

$$\Pr(\text{error} \mid \mathbf{x}) = \frac{\min_k \Pr(\mathbf{x} \mid y_k) \Pr(y_k)}{\Pr(\mathbf{x})} \quad (1.1.23.1)$$

Solution: The classifier selects the class with the maximum posterior probability. The error probability is determined by the unselected (minimum) posterior:

$$\min \left\{ \frac{3}{4} \cdot \frac{1}{3}, \frac{1}{4} \cdot \frac{2}{3} \right\} = \frac{1}{6} \quad (1.1.23.2)$$

$$\Pr(\mathbf{x}) = \frac{1}{4} + \frac{1}{6} = \frac{5}{12} \quad (1.1.23.3)$$

$$\Pr(\text{error} \mid \mathbf{x}) = \frac{1/6}{5/12} = 0.40 \quad (1.1.23.4)$$

1.1.24 Moments of Standard Normal and Chi-Square

$$Z \sim \mathcal{N}(0, 1) \implies E[Z^2] = 1, E[Z^4] = 3 \quad (1.1.24.1)$$

Solution: For $Y = Z^2$:

$$\text{Var}(Y) = E[Y^2] - (E[Y])^2 = E[Z^4] - (E[Z^2])^2 = 3 - 1^2 = 2 \quad (1.1.24.2)$$

1.1.25 Rank of Involutory Matrices

$$\mathbf{A}^3 = \mathbf{A} \implies \text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}^2) \quad (1.1.25.1)$$

Solution: Factoring $\mathbf{A}^3 - \mathbf{A} = \mathbf{A}(\mathbf{A} - \mathbf{I})(\mathbf{A} + \mathbf{I}) = \mathbf{0}$. The minimal polynomial has distinct linear roots, making $\mathbf{A}$ diagonalizable. The Rank-Nullity theorem confirms equal rank across powers.

1.1.26 Gram Matrix Positive Definiteness

$$\mathbf{A} = \mathbf{X}^\top \mathbf{X} \implies \mathbf{z}^\top \mathbf{A} \mathbf{z} = \|\mathbf{X} \mathbf{z}\|^2 \quad (1.1.26.1)$$

Solution: If the columns of $\mathbf{X}$ are linearly independent, $\mathbf{X}\mathbf{z} = \mathbf{0} \iff \mathbf{z} = \mathbf{0}$. Thus, $\|\mathbf{X}\mathbf{z}\|^2 > 0$ for all non-zero $\mathbf{z}$, rendering $\mathbf{A}$ positive definite and invertible.

1.1.27 Interval Probabilities from CDF

$$\Pr(X^2 \leq c^2) = F_X(c) - F_X(-c) \quad (1.1.27.1)$$

Solution: For $F_X(x) = \frac{1}{4}(x+1)^2$:

$$\Pr(X^2 \leq 0.25) = F_X(0.5) - F_X(-0.5) = \frac{2.25}{4} - \frac{0.25}{4} = 0.5 \quad (1.1.27.2)$$

1.1.28 Central Limit Theorem for Binomial Sums

$$Z_n = \frac{\sum X_i - n\theta}{\sqrt{n\theta(1-\theta)}} \xrightarrow{d} \mathcal{N}(0, 1) \quad (1.1.28.1)$$

Solution: For $n = 300$, $\theta = 0.25$, $\mu = 75$ and $\sigma = 7.5$:

$$\Pr(60 \leq Y \leq 90) \approx \phi\left(\frac{90 - 75}{7.5}\right) - \phi\left(\frac{60 - 75}{7.5}\right) = \phi(2) - \phi(-2) \quad (1.1.28.2)$$

1.1.29 Geometric Discretization of Exponential

$$\Pr(\lfloor X \rfloor = l) = e^{-\lambda l} (1 - e^{-\lambda}) \quad (1.1.29.1)$$

Solution: Integrating $\lambda e^{-\lambda x}$ over $[l, l+1]$ yields $q^l(1-q)$ with $q = e^{-\lambda}$.

1.1.30 Complement Rule

$$\Pr(\text{at least one success}) = 1 - (1-p)^n \quad (1.1.30.1)$$

Solution: The probability of rolling zero 1s on 100 fair dice is $(5/6)^{100}$.

1.1.31 Critical Points

$$f'(x) = 0 \text{ at local extrema} \quad (1.1.31.1)$$

Solution: Local extrema require evaluating roots of $f'(x)$ and checking $f''(x)$ convexity. Extrema of the derivative itself locate inflection points.

1.1.32 Orthogonal Projection Matrices

$$\mathbf{A} = \sum_{i=1}^k \mathbf{x}_i \mathbf{x}_i^\top \implies \mathbf{A}^2 = \mathbf{A} \quad (1.1.32.1)$$

Solution: Using orthonormal properties, $\mathbf{A}$ is idempotent. Singular values equal eigenvalues $\sigma \in \{0, 1\}$. For incomplete bases $k < n$, $\mathbf{A}$ is singular.

1.1.33 Strict Convexity

$$f''(x) > 0 \implies \text{Unique global minimum (if any)} \quad (1.1.33.1)$$

Solution: A strictly monotonic $f'(x)$ cannot possess multiple roots, precluding multiple local minima.

1.1.34 Isometric Norm Preservation

$$\|\mathbf{A}\mathbf{x}\|^2 = \|\mathbf{x}\|^2 \iff \mathbf{A}^\top \mathbf{A} = \mathbf{I} \quad (1.1.34.1)$$

Solution: Applying polarization identity, the equality strictly implies orthogonality and full rank.

1.1.35 SVM Margin

$$\gamma = \frac{2}{\|\mathbf{w}\|} \quad (1.1.35.1)$$

Solution: Support vectors lie directly on the canonical hyperplanes $y_i (\mathbf{w}^\top \mathbf{x}_i + b) = 1$. The geometric width is inverse to norm $\mathbf{w}$.

1.1.36 Variance of Sample Mean

$$\text{Var} \left(\frac{1}{n} \sum_{i=1}^n X_i \right) = \frac{\sigma^2}{n} \quad (1.1.36.1)$$

Solution: Variance of a sum of independent identical variables scales by n , rendering the sample mean variance inversely proportional to n .

1.1.37 Shortest Path Metric

$$d(u, v) \leq d(u, w) + d(w, v) \quad (1.1.37.1)$$

Solution: Fake edges violating the triangle inequality are identifiable by calculating the minimum spanning distances directly.

1.1.38 Lipschitz Continuity with Exponent > 1

$$|f(x) - f(y)| \leq K |x - y|^\alpha \implies f'(x) = 0 \quad (1.1.38.1)$$

Solution: Dividing by $|x - y|$ and taking the limit $y \rightarrow x$ for $\alpha > 1$ yields a derivative of zero, enforcing a constant function.

1.1.39 Rayleigh-Ritz Theorem for PCA

$$\max_{\|\mathbf{u}\|=1} \mathbf{u}^\top \mathbf{S} \mathbf{u} = \lambda_{\max}(\mathbf{S}) \quad (1.1.39.1)$$

Solution: Empirical variance along $\mathbf{u}$ is maximized by the eigenvector corresponding to the principal eigenvalue.

1.1.40 Expectation of Indicator Sums

$$E \left[\sum_{i=1}^n I_i \right] = \sum_{i=1}^n E[I_i] = np \quad (1.1.40.1)$$

Solution: Linearity of expectation holds uniformly irrespective of variable dependence or independence.

1.1.41 Relational Division

$$R \div S = \pi_A(R) - \pi_A((\pi_A(R) \times S) - R) \quad (1.1.41.1)$$

Solution: Simulates universal quantification $\forall$ in relational schema.

B. Problems

- 1.2.1 A 4×4 digital image has pixel intensities (U) as shown in the figure. The number of pixels with $U \leq 4$ is: (DA 2025)

0	1	0	2
4	7	3	3
5	5	4	4
6	7	3	2

Fig. 1.2.1.1: Pixel Grid

- a) 3 b) 8 c) 11 d) 9

1.2.2 A rectangle has a length L and a width W , where $L > W$. If the width, W , is increased by 10%, which one of the following statements is correct for all values of L and W ? (DA 2025)

- a) Perimeter increases by 10%.
b) Length of the diagonals increases by 10%.
c) Area increases by 10%.
d) The rectangle becomes a square.

1.2.3 Weight of a person can be expressed as a function of their age. The function usually varies from person to person. Suppose this function is identical for two brothers, and it monotonically increases till the age of 50 years and then it monotonically decreases. Let a_1 and a_2 (in years) denote the ages of the brothers and $a_1 < a_2$. Which one of the following statements is correct about their age on the day when they attain the same weight? (DA 2025)

- a) $a_1 < a_2 < 50$ c) $50 < a_1 < a_2$
b) $a_1 < 50 < a_2$ d) Either $a_1 = 50$ or $a_2 = 50$

1.2.4 If a real variable x satisfies $3^{x^2} = 27 \times 9^x$, then the value of $\frac{2^{x^2}}{(2^x)^2}$ is: (DA 2025)

- a) 2^{-1} b) 2^0 c) 2^3 d) 2^{15}

1.2.5 Suppose X and Y are random variables. The conditional expectation of X given Y is denoted by $E[X | Y]$. Then $E[E[X | Y]]$ equals (DA 2025)

- a) $E[X | Y]$ b) $\frac{E[X]}{E[Y]}$ c) $E[X]$ d) $E[Y]$

1.2.6 The number of additions and multiplications involved in performing Gaussian elimination on any $n \times n$ upper triangular matrix is of the order (DA 2025)

- a) $O(n)$ b) $O(n^2)$ c) $O(n^3)$ d) $O(n^4)$

1.2.7 The sum of the elements in each row of $\mathbf{A} \in \mathbb{R}^{n \times n}$ is 1. If $\mathbf{B} = \mathbf{A}^3 - 2\mathbf{A}^2 + \mathbf{A}$, which one of the following statements is correct (for $\mathbf{x} \in \mathbb{R}^n$)? (DA 2025)

- a) The equation $\mathbf{B}\mathbf{x} = \mathbf{0}$ has no solution
 b) The equation $\mathbf{B}\mathbf{x} = \mathbf{0}$ has exactly two solutions
 c) The equation $\mathbf{B}\mathbf{x} = \mathbf{0}$ has infinitely many solutions
 d) The equation $\mathbf{B}\mathbf{x} = \mathbf{0}$ has a unique solution

1.2.8 Let $f(x) = \frac{e^x - e^{-x}}{2}$, $x \in \mathbb{R}$. Let $f^{(k)}(a)$ denote the k^{th} derivative of f evaluated at a . What is the value of $f^{(10)}(0)$? (Note: ! denotes factorial) (DA 2025)

- a) 0 b) 1 c) $\frac{1}{10!}$ d) $\frac{2}{10!}$

1.2.9 Consider the following three relations:

Car (model, year, serial, color)

Make (maker, model)

Own (owner, serial)

A tuple in Car represents a specific car of a given model, made in a given year, with a serial number and a color. A tuple in Make specifies that a maker company makes cars of a certain model. A tuple in Own specifies that an owner owns the car with a given serial number. Keys are underlined; (owner, serial) together form key for Own. ($\bowtie$ denotes natural join)

$$\pi_{\text{owner}}(\text{Own} \bowtie (\sigma_{\text{color}='red'}(\text{Car} \bowtie (\sigma_{\text{maker}='ABC'}(\text{Make})))))) \quad (1.2.9.1)$$

Which one of the following options describes what the above expression computes? (DA 2025)

- a) All owners of a red car, a car made by ABC, or a red car made by ABC
 b) All owners of more than one car, where at least one car is red and made by ABC
 c) All owners of a red car made by ABC
 d) All red cars made by ABC

1.2.10 Consider a hash table of size 10 with indices $\{0, 1, \dots, 9\}$, with the hash function

$$h(x) = 3x \pmod{10} \quad (1.2.10.1)$$

where linear probing is used to handle collisions. The hash table is initially empty and then the following sequence of keys is inserted into the hash table: 1, 4, 5, 6, 14, 15. The indices where the keys 14 and 15 are stored are, respectively (DA 2025)

- a) 2 and 5 b) 2 and 6 c) 4 and 5 d) 4 and 6

1.2.11 Let X be a continuous random variable whose cumulative distribution function (CDF)

$F_X(x)$, for some t , is given as follows:

$$F_X(x) = \begin{cases} 0 & x \leq t \\ \frac{x-t}{4-t} & t \leq x \leq 4 \\ 1 & x \geq 4 \end{cases} \quad (1.2.11.1)$$

If the median of X is 3, then what is the value of t ? (DA 2025)

- a) 2 b) 1 c) -1 d) 0

1.2.12 Let $X = aZ + b$, where Z is a standard normal random variable, and a, b are two unknown constants. It is given that

$$E[X] = 1, \quad E[(X - E[X])Z] = -2, \quad E[(X - E[X])^2] = 4, \quad (1.2.12.1)$$

where $E[X]$ denotes the expectation of random variable X . The values of a, b are: (DA 2025)

- a) $a = -2, b = 1$ c) $a = -2, b = -1$
b) $a = 2, b = -1$ d) $a = 1, b = 1$

1.2.13 It is given that $P(X \geq 2) = 0.25$ for an exponentially distributed random variable X with $E[X] = \frac{1}{\lambda}$, where $E[X]$ denotes the expectation of X . What is the value of λ ? ($\ln$ denotes natural logarithm) (DA 2025)

- a) $\ln 2$ b) $\ln 4$ c) $\ln 3$ d) $\ln 0.25$

1.2.14 Consider the following Python declarations of two lists.

```
A = [1, 2, 3]
B = [4, 5, 6]
```

Which one of the following statements results in $A = [1, 2, 3, 4, 5, 6]$? (DA 2025)

- a) `A.extend(B)` b) `A.append(B)` c) `A.update(B)` d) `A.insert(B)`

1.2.15 Consider two functions $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow (1, \infty)$. Both functions are differentiable at a point c . Which of the following functions is/are ALWAYS differentiable at c ? The symbol $\cdot$ denotes product and the symbol $\circ$ denotes composition of functions. (DA 2025)

- a) $f \pm g$ b) $f \cdot g$ c) $\frac{f}{g}$ d) $f \circ g + g \circ f$

1.2.16 Which of the following statements is/are correct? (DA 2025)

- a) $\mathbb{R}^n$ has a unique set of orthonormal basis vectors
b) $\mathbb{R}^n$ does not have a unique set of orthonormal basis vectors

c) Linearly independent vectors in $\mathbb{R}^n$ are orthonormal

d) Orthonormal vectors in $\mathbb{R}^n$ are linearly independent

1.2.17 For which of the following inputs does binary search take time $O(\log n)$ in the worst case? (DA 2025)

a) An array of n integers in any order

b) A linked list of n integers in any order

c) An array of n integers in increasing order

d) A linked list of n integers in increasing order

1.2.18 Let $\mathbf{A} = \mathbf{I}_n + \mathbf{x}\mathbf{x}^\top$ where $\mathbf{I}_n$ is the $n \times n$ identity matrix and $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{x}^\top \mathbf{x} = 1$. Which of the following options is/are correct? (DA 2025)

a) Rank of $\mathbf{A}$ is n

b) $\mathbf{A}$ is invertible

c) 0 is an eigenvalue of $\mathbf{A}$

d) $\mathbf{A}^{-1}$ has a negative eigenvalue

1.2.19 Suppose that insertion sort is applied to the array $[1, 3, 5, 7, 9, 11, x, 15, 13]$ and it takes exactly two swaps to sort the array. Select all possible values of x . (DA 2025)

a) 10

b) 12

c) 14

d) 16

1.2.20 There are three boxes containing white balls and black balls.

- Box-1 contains 2 black and 1 white balls.
- Box-2 contains 1 black and 2 white balls.
- Box-3 contains 3 black and 3 white balls.

In a random experiment, one of these boxes is selected, where the probability of choosing Box-1 is $\frac{1}{2}$, Box-2 is $\frac{1}{6}$, and Box-3 is $\frac{1}{3}$. A ball is drawn at random from the selected box. Given that the ball drawn is white, the probability that it is drawn from Box-2 is _____.

(DA 2025)

1.2.21 $\lim_{t \rightarrow +\infty} \sqrt{t^2 + t} - t = \underline{\hspace{2cm}}$.

(DA 2025)

1.2.22 On a relation named Loan of a bank:

loan_number	branch_name	amount
L11	Banjara Hills	90000
L14	Kondapur	50000
L15	SR Nagar	40000
L22	SR Nagar	25000
L23	Balanagar	80000
L25	Kondapur	70000
L19	SR Nagar	65000

Fig. 1.2.22.1

the following SQL query is executed.

```
SELECT L1.loan_number
```

```
FROM Loan L1
WHERE L1.amount > (SELECT MAX (L2.amount)
FROM Loan L2
WHERE L2.branch_name = 'SR Nagar');
```

The number of rows returned by the query is _____. (DA 2025)

- 1.2.23 Given data $\{(-1, 1), (2, 5), (3, 5)\}$ of the form (x, y) , we fit a model $y = wx$ using linear least-squares regression. The optimal value of w is _____. (DA 2025)
- 1.2.24 The naive Bayes classifier is used to solve a two-class classification problem with class-labels y_1, y_2 . Suppose the prior probabilities are $P(y_1) = \frac{1}{3}$ and $P(y_2) = \frac{2}{3}$. Assuming a discrete feature space with

$$P(\mathbf{x} | y_1) = \frac{3}{4} \quad \text{and} \quad P(\mathbf{x} | y_2) = \frac{1}{4} \quad (1.2.24.1)$$

for a specific feature vector $\mathbf{x}$. The probability of misclassifying $\mathbf{x}$ is _____. (DA 2025)

- 1.2.25 Let $Y = Z^2$, $Z = \frac{X - \mu}{\sigma}$, where X is a normal random variable with mean μ and variance σ^2 . The variance of Y is (DA 2025)
- a) 1 b) 2 c) 3 d) 4

- 1.2.26 Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be such that $\mathbf{A}^3 = \mathbf{A}$. Which one of the following statements is ALWAYS correct? (DA 2025)
- a) $\mathbf{A}$ is invertible
b) Determinant of $\mathbf{A}$ is 0
c) The sum of the diagonal elements of $\mathbf{A}$ is 1
d) $\mathbf{A}$ and $\mathbf{A}^2$ have the same rank

- 1.2.27 Let $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ be a set of linearly independent vectors in $\mathbb{R}^n$. Let the (i, j) -th element of matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ be given by $A_{ij} = \mathbf{x}_i^\top \mathbf{x}_j$, $1 \leq i, j \leq n$. Which one of the following statements is correct? (DA 2025)
- a) $\mathbf{A}$ is invertible
b) 0 is a singular value of $\mathbf{A}$
c) Determinant of $\mathbf{A}$ is 0
d) $\mathbf{z}^\top \mathbf{A} \mathbf{z} = 0$ for some non-zero $\mathbf{z} \in \mathbb{R}^n$

- 1.2.28 Consider the cumulative distribution function (CDF) of a random variable X :

$$F_X(x) = \begin{cases} 0 & x \leq -1 \\ \frac{1}{4}(x+1)^2 & -1 \leq x \leq 1 \\ 1 & x \geq 1 \end{cases} \quad (1.2.28.1)$$

The value of $P(X^2 \leq 0.25)$ is (DA 2025)

- a) 0.625 b) 0.25 c) 0.5 d) 0.5625
- 1.2.29 A random variable X is said to be distributed as Bernoulli(θ), denoted by $X \sim \text{Bernoulli}(\theta)$, if $P(X = 1) = \theta$, $P(X = 0) = 1 - \theta$ for $0 < \theta < 1$. Let $Y =$

Which of the following statements is/are correct?

(DA 2025)

- a) The maximum value of f is attained at $x = -5$
- b) The minimum value of f is attained at $x = -2$
- c) The maximum value of f is $\frac{133}{2}$
- d) The minimum value of the derivative of f is attained at $x = -\frac{7}{2}$

1.2.34 Let $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4, \mathbf{x}_5$ be a system of orthonormal vectors in $\mathbb{R}^{10}$. Consider the matrix $\mathbf{A} = \mathbf{x}_1\mathbf{x}_1^\top + \cdots + \mathbf{x}_5\mathbf{x}_5^\top$. Which of the following statements is/are correct? (DA 2025)

- a) Singular values of $\mathbf{A}$ are also its eigenvalues
- b) Singular values of $\mathbf{A}$ are either 0 or 1
- c) Determinant of $\mathbf{A}$ is 1
- d) $\mathbf{A}$ is invertible

1.2.35 Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a twice-differentiable function and suppose its second derivative satisfies $f''(x) > 0$ for all $x \in \mathbb{R}$. Which of the following statements is/are ALWAYS correct? (DA 2025)

- a) f has a local minima
- b) There does not exist x and y , $x \neq y$ such that $f'(x) = f'(y) = 0$
- c) f has at most one global minimum
- d) f has at most one local minimum

1.2.36 An $n \times n$ matrix $\mathbf{A}$ with real entries satisfies the property: $\|\mathbf{A}\mathbf{x}\|^2 = \|\mathbf{x}\|^2$, for all $\mathbf{x} \in \mathbb{R}^n$, where $\|\cdot\|$ denotes the Euclidean norm. Which of the following statements is/are ALWAYS correct? (DA 2025)

- a) $\mathbf{A}$ must be orthogonal
- b) $\mathbf{A} = \mathbf{I}$ where $\mathbf{I}$ denotes the identity matrix, is the only solution
- c) The eigenvalues of $\mathbf{A}$ are either +1 or -1
- d) $\mathbf{A}$ has full rank

1.2.37 Consider designing a linear binary classifier $f(\mathbf{x}) = \text{sign}(\mathbf{w}^\top \mathbf{x} + b)$, $\mathbf{x} \in \mathbb{R}^2$ on the following training data:

$$\text{Class-1 : } \left\{ \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 2 \end{pmatrix} \right\}, \quad \text{Class-2 : } \left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\} \quad (1.2.37.1)$$

Hard-margin support vector machine (SVM) formulation is solved to obtain $\mathbf{w}$ and b . Which of the following options is/are correct? (DA 2025)

- a) $\mathbf{w} = \begin{pmatrix} 4 \\ 4 \end{pmatrix}$ and $b = 1$
- b) The number of support vectors is 3
- c) The margin is $\sqrt{2}$
- d) Training accuracy is 98%

1.2.38 Consider a coin-toss experiment where the probability of head showing up is p . In the i^{th} coin toss, let $X_i = 1$ if head appears, and $X_i = 0$ if tail appears. Consider

$$\hat{p} = \frac{1}{n} \sum_{i=1}^n X_i \quad (1.2.38.1)$$

where n is the total number of independent coin tosses. Which of the following

statements is/are correct?

(DA 2025)

- a) $E[\hat{p}] = p$
- b) $E[\hat{p}] = \frac{p}{n}$
- c) As n increases, variance of $\hat{p}$ decreases
- d) Variance of $\hat{p}$ does not depend on n

1.2.39 Let G be a simple, unweighted, and undirected graph. A subset of the vertices and edges of G are shown below.

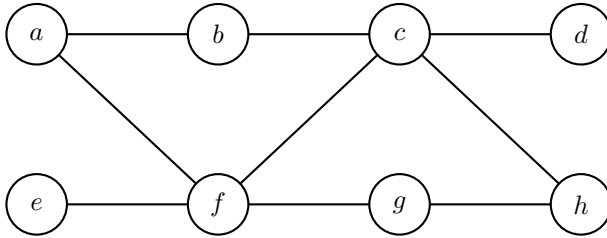


Fig. 1.2.39.1

It is given that $a - b - c - d$ is a shortest path between a and d ; $e - f - g - h$ is a shortest path between e and h ; $a - f - c - h$ is a shortest path between a and h . Which of the following is/are NOT the edges of G ? (DA 2025)

- a) (b, d)
- b) (b, g)
- c) (b, h)
- d) (e, g)

1.2.40 Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be such that $|f(x) - f(y)| \leq (x - y)^2$ for all $x, y \in \mathbb{R}$. Then

$$f(1) - f(0) = \underline{\hspace{2cm}} \quad (1.2.40.1)$$

(DA 2025)

1.2.41 Let $D = \{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(n)}\}$ be a dataset of n observations where each $\mathbf{x}^{(i)} \in \mathbb{R}^{100}$. It is given that $\sum_{i=1}^n \mathbf{x}^{(i)} = \mathbf{0}$. The covariance matrix computed from D has eigenvalues $\lambda_i = 100^{2-i}$, $1 \leq i \leq 100$. Let $\mathbf{u} \in \mathbb{R}^{100}$ be the direction of maximum variance with $\mathbf{u}^\top \mathbf{u} = 1$. The value of

$$\frac{1}{n} \sum_{i=1}^n \left(\mathbf{u}^\top \mathbf{x}^{(i)} \right)^2 = \underline{\hspace{2cm}} \quad (1.2.41.1)$$

(DA 2025)

1.2.42 A bag contains 5 white balls and 10 black balls. In a random experiment, n balls are drawn from the bag one at a time with replacement. Let S_n denote the total number of black balls drawn in the experiment. The expectation of S_{100} denoted by $E[S_{100}] = \underline{\hspace{2cm}}$. (DA 2025)

1.2.43 Consider the following tables, Loan and Borrower, of a bank.

Loan

loan_num	branch_name	amount
L11	Banjara Hills	90000
L14	Kondapur	50000
L15	SR Nagar	40000
L22	SR Nagar	25000
L23	Balanagar	80000
L25	Kondapur	70000
L19	SR Nagar	65000

Borrower

customer_name	loan_num
Anand	L11
Karteek	L11
Karteek	L14
Ankita	L15
Gopal	L19
Karteek	L22
Karteek	L23
Sunil	L23
Sunil	L25

Fig. 1.2.43.1

Query: $\pi_{\text{branch_name}, \text{customer_name}}(\text{Loan} \bowtie \text{Borrower}) \div \pi_{\text{branch_name}}(\text{Loan})$ where $\bowtie$ denotes natural join. The number of tuples returned by the above relational algebra query is _____. (DA 2025)

1.2.44 Consider the following Python code snippet.

```
def f(a, b):
    if (a == 0):
        return b
    if (a % 2 == 1):
        return 2 * f((a - 1) / 2, b)
    return b + f(a - 1, b)

print(f(15, 10))
```

The value printed by the code snippet is _____. (DA 2025)

1.2.45 Consider the following pseudocode.

```
Create empty stack S
Set x = 0, flag = 0, sum = 0
Push x onto S

while (S is not empty) {
    if (flag equals 0){
        Set x = x + 1
        Push x onto S
    }
    if (x equals 8):
        Set flag = 1
        if (flag equals 1){
            x = Pop(S)
            if (x is odd):
```

	Pop(S)
	Set sum = sum + x
	}
}	
Output sum	

The value of sum output by a program executing the above pseudocode is _____.
(DA 2025)

- 1.2.46 Consider a directed graph $G = (V, E)$, where $V = \{0, 1, 2, \dots, 100\}$ and $E = \{(i, j) : 0 < j - i \leq 2 \text{ for all } i, j \in V\}$. Suppose the adjacency list of each vertex is in decreasing order of vertex number, and depth-first search (DFS) is performed at vertex 0. The number of vertices that will be discovered after vertex 50 is _____.
(DA 2025)